FDS EXPERIMENT 20

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Experiment to understand K means clustering algorithm for a given data set.

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
```

```
data = {
    "Study_Hours": [1, 2, 1.5, 5, 6, 5.5, 8, 9, 8.5],
    "Marks": [35, 40, 38, 70, 75, 72, 90, 95, 92]
}

df = pd.DataFrame(data)

print(df)
```

```
   Study_Hours  Marks
0          1.0     35
1          2.0     40
2          1.5     38
3          5.0     70
4          6.0     75
5          5.5     72
6          8.0     90
7          9.0     95
8          8.5     92
```

```python
X = df[["Study_Hours", "Marks"]]

kmeans = KMeans(
    n_clusters=3,
    random_state=42,
    n_init=10
)

kmeans.fit(X)

df["Cluster"] = kmeans.labels_

print(df)
```

```
    Study_Hours  Marks  Cluster
0           1.0     35        1
1           2.0     40        1
2           1.5     38        1
3           5.0     70        0
4           6.0     75        0
5           5.5     72        0
6           8.0     90        2
7           9.0     95        2
8           8.5     92        2
```

```
print("Cluster Centers:")
print(kmeans.cluster_centers_)
```

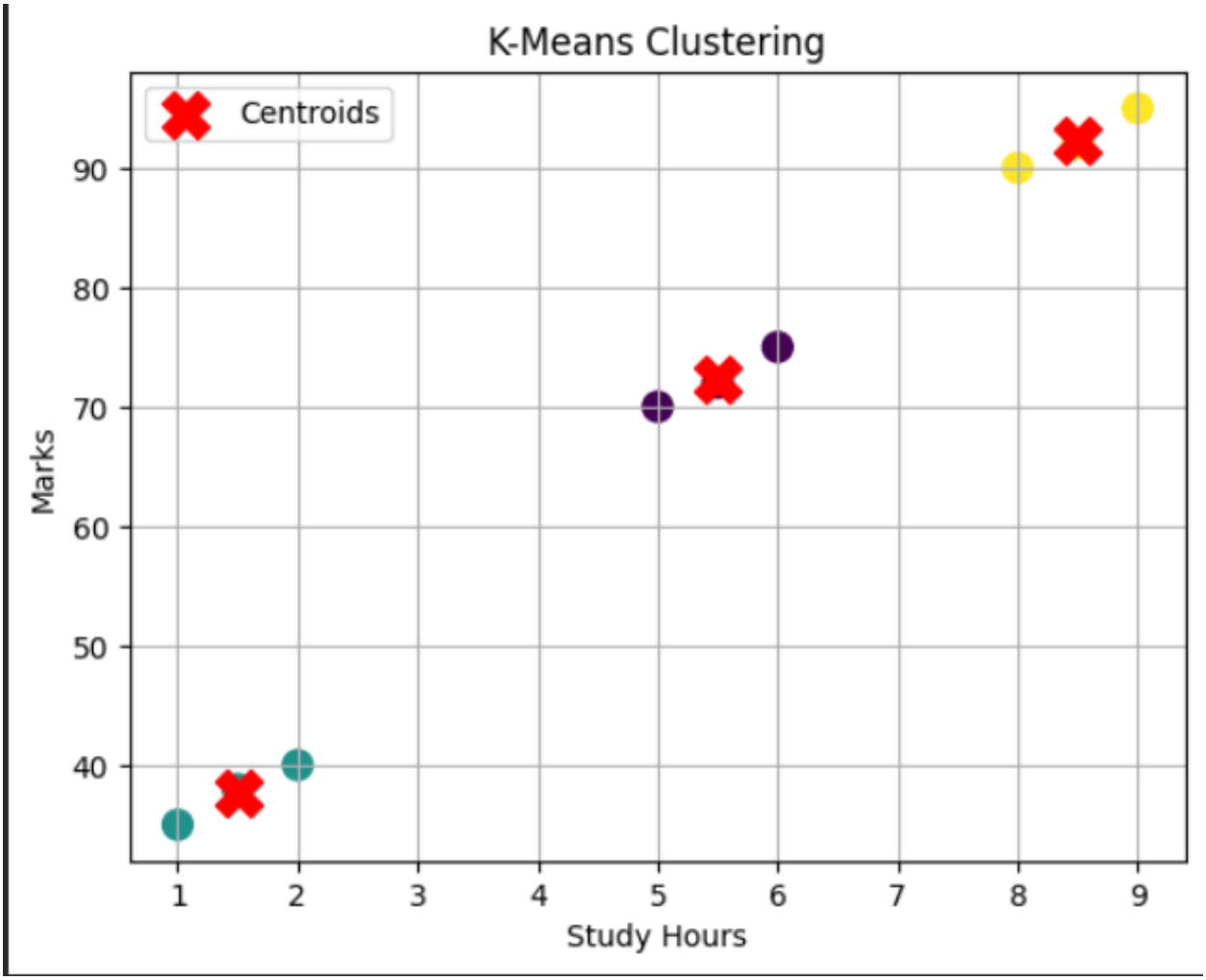
```
Cluster Centers:
[[ 5.5        72.33333333]
 [ 1.5        37.66666667]
 [ 8.5        92.33333333]]
```

```
plt.scatter(
    df["Study_Hours"],
    df["Marks"],
    c=df["Cluster"],
    cmap="viridis",
    s=100
)

centers = kmeans.cluster_centers_

plt.scatter(
    centers[:, 0],
    centers[:, 1],
    color="red",
    marker="X",
    s=250,
    label="Centroids"
)

plt.xlabel("Study Hours")
plt.ylabel("Marks")
plt.title("K-Means Clustering")
plt.legend()
plt.grid(True)
plt.show()
```


K-Means Clustering
Centroids
Marks
Study Hours
90
80
70
60
50
40
1
2
3
4
5
6
7
8
9